

Chapter three Data classification and presentation

Tabulation:

Tabulation is the logical listing of related quantitative data in vertical column and horizontal rows of numbers with sufficient explanatory and qualifying words, phrases and statements in the form of titles, headings and notes to make clear the full meaning of data and their origin.

Table is a systematic arrangement of statistical data in columns and rows. Parts of a table;

- 1) Table Number, 2) Title of the table, 3) Caption – column heading, 4) Stub – row heading, 5) Body – Numerical information, 6) Foot Notes – to classify anything in the table.

Annual Income & Expenditure of a Business Farm

Year	1999	2000	2001	2002	2003	2004
Income (In Million Taka)	12	14	15	17	22	30
Expenditure (In Million Taka)	08	11	13	16	21	29.5

*** The annual report of the Business farm during the period 1999 – 2004**

Frequency:

Frequency:

It is the number of occurrences of any individual values in set of data or observations. It is denoted by f_i . For Example, The marks of 10 students are 50,30,50,40,45,60,50,30,45,50 in a set of data. Then we get,

Marks	Frequency (f_i)
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30	2
40	1
45	2
50	4
60	1

	$\sum f_i = 10 = (N)$

Frequency distribution:

A Frequency distribution refers to data classified on the basis of some variable that can be measured, such as wages, ages, prices, and number of units produced.

In other word, a tabular arrangement of data by classes together with the corresponding class frequencies is called a frequency distribution.

Variable:

The changeable properties of data are called variable. The term 'Variable' refers to the characteristic that varies in amount or magnitude in a frequency distribution.

Types of Variable:

- i) **Qualitative Variable** (measured in figure): Gender, Honesty, Colour, Morality
- ii) **Quantitative Variable** (not measured in figure): Price, Production, Cost

Types of Quantitative Variable

- I) **Discrete Variable** (Full digit / integer): Family member, Goals
- II) **Continuous Variable** (lies between two limits) : Income, Age

Construction of a frequency distribution:

Following are the steps for the construction of a frequency distribution:

- i) Find out range by subtracting the lowest value from the highest value of the variable X.
- ii) The number of classes should not be too large or too small, usually it lies, between 5 to 20 considering the practical situation. Alternative way to determine no. of class, $K = 1 + 3.322 \log n$.
- iii) Having fixed the number of classes divide the range by it and the nearest integer to this value gives length of class intervals.
- iv) A frequency distribution will have mainly three columns having names, **class interval**, **tally marks** and **frequency**. The first class interval will start with the smallest value and continue until interval with the highest value of the given series of data is reached.
- v) There are two methods of classifying the data according to class intervals, namely
 - a) Exclusive method.
 - b) Inclusive method.

a) Exclusive method:

When the class intervals are so defined that the upper limit of one class is lower limit of the next class, it is known as the exclusive method of classification. The following are classified on this basis;

Incomes(Tk.)	number of employees
800 – 900	50
900 – 1000	100
1000 – 1100	200

b) Inclusive method:

Under the Inclusive method of classification the upper limit of one class is included in that class itself. The following example illustrates the method;

Incomes(Tk.)	number of employees
800 – 899	50

900 – 999	100
1000 – 1099	200

- vi) Give tick mark to each of the values of the original table of raw data and put tally mark against the appropriate class interval. Thus exhaust all the values one after another.
- vii) Count the number of tally marks corresponding to each class interval and write the result in the respective frequency column.

Mid Point:

It is obtained by adding the lower and upper class limits and dividing by 2. Thus the class mark of the class interval 80-62 is $(60+62)/2 = 61$. For purposes of further mathematical analysis, all observations belonging to a given class interval are assumed to coincide with the class mark. Thus all weight in the class interval 60 – 62 kg are considered to be 61 kg.

Class intervals and class limits:

A symbol defining a class, such as 60 – 62 kg, is called a class interval. The end numbers, 60 & 62, are called class limits; the smaller number (60) is the lower class limit and the larger number (62) is the upper class limit.

A class interval that at least theoretically, has either no upper class limit or no lower class limit indicated is called an open class interval. For example, referring to age groups of individuals, the class interval “65 and over ” is an open class interval.

Example 01:

Form a frequency distribution taking a suitable class interval for the following data giving the ages of 52 employees in a governmental agency.

67 34 36 48 49 31 61 34 43 45 38 32 27 61 29 47 36 50 46 30 46 32 30 33 45 49 48 41 53
36 37 37 47 30 46 50 28 35 35 38 36 46 43 34 62 69 50 28 44 43 60 39

Solution:

Here,

The largest age = 69 years

The smallest age = 27 years

The range = $69 - 27 = 42$ years

If 5 classes are made, the class interval size is $42/5 = 8.4$ it is equivalent to 8

Now, taking class interval size is 8 and following inclusive method of classification, a frequency distribution is given below:

Table 01: Frequency distribution of ages of 52 employees

Class interval of age (in years)	Tally	Frequency (No. of employees)	Percentage of employees
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25-32			
33-40			
41-48			
49-56			
57-64			
65-72			
Total			

Again, taking class interval size is 8 and following exclusive method of classification, the frequency distribution is given below:

Frequency distribution of ages of 52 employees

Class interval of age (in years)	Tally	Frequency
26 – 34		
34 – 42	I	
42 – 50		
50 – 58		
58 – 66		
66 – 74	I	
Total		

Exercise 01:

The daily sale (in thousand tk.) of 30 shops of a market is given below:

19, 56, 90, 36, 58, 25, 58, 56, 50, 63, 70, 82, 94, 23, 78

51, 57, 42, 39, 75, 83, 79, 51, 61, 87, 26, 49, 37, 88, 37

Construct a frequency distribution taking suitable class interval using i) inclusive method; ii) exclusive method.

Solution:

Here, Highest sale=Tk. 94 thousand

Lowest sale= 19

Range=94-19=75

If 5 classes are made, the class interval size is $75/5 = 15$

If 20 classes are made, the class interval size is $75/20 = 3.75$

Now, taking class interval size is 15 and following inclusive method of classification, a frequency distribution is given below:

Table 01: Frequency distribution of daily sales of 30 shops

Class interval of daily sale (in 000 Tk.)	Tally	Frequency (No. of shops)	Percentage of shops
15-29			
30-44			
45-59			
60-74			
75-89			
90-104			
Total			

Exercise 02:

The price (in Tk.) of 40 books of a bookshop is given below:

140, 500, 550, 100, 900, 960, 670, 291, 657, 239
651, 805, 465, 369, 741, 296, 652, 536, 582, 591
891, 788, 736, 908, 391, 458, 194, 38, 165, 591
519, 367, 491, 102, 297, 561, 518, 861, 725, 493

Construct a frequency distribution using suitable class interval.

Solution:

Exercise 03:

The Bangladesh Transportation Commission is concerned about the speed motorists driving on a section of the main highway. Here are the speeds of 45 motorists:

15 32 45 46 42 39 68 47 18 31 48 49 56 52 39 48 69 61 44 42 38 52
55 58 62 58 48 56 55 49 58 48 47 52 37 64 29 55 38 29 62 49 69 18
61.

Construct a frequency distribution using suitable class interval.

Exercise 04:

The profits (in lakhs) of 30 companies for the year 2001-2002 are given below:

20, 22, 35, 42, 37, 42, 48, 53, 49, 65, 39, 48, 67, 18, 16
23, 37, 35, 49, 63, 65, 55, 45, 58, 57, 69, 25, 29, 58, 65.

Construct a frequency distribution taking a suitable class-interval and draw a cumulative frequency curve.

Representation of Data:

Data can be presented by two ways:

- a) By graph
- b) By Diagram

A frequency distribution can be represented graphically in any of the following ways:

1. Histogram. 2. Frequency polygon 3. Frequency curve.
4. Cumulative frequency curve or Ogive.

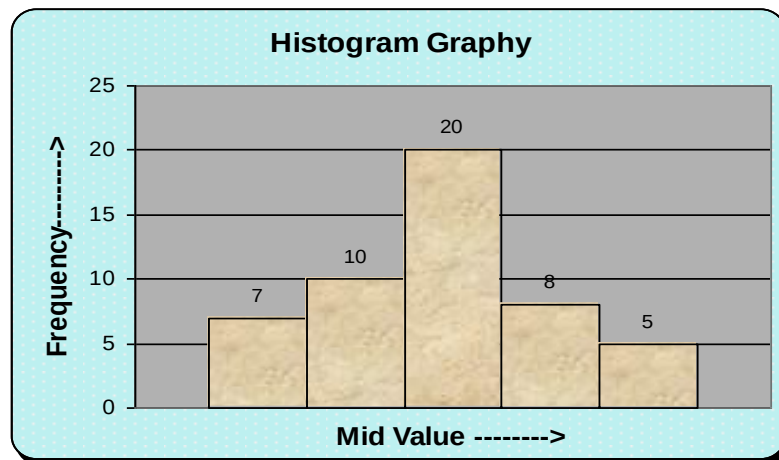
Histogram:

The statistical meaning of histogram is that it is a graph that represents the class frequencies in a frequency distribution by vertical adjacent rectangles. A histogram is a graphical method for presenting the data, where the observations are located on a horizontal axis and the frequency of those observations is depicted along the vertical axis.

Example 02:

Draw Histogram for the following frequency distribution of marks of 50 students.

Marks	0-20	20-40	40-60	60-80	80-100	Total
No. of students(f)	7	10	20	8	5	50



Frequency

In frequency polygon the mid-values of continuous class interval are represented along X-axis and the frequencies corresponding to the class intervals are presented along the Y-axis. The class frequencies are plotted against the mid-values of the respective class intervals. The points are then joint by straight lines one after another. The first and last points are then brought down at each end to the axis by joining it to the mid-values of the next out lying interval of zero frequency. The polygon thus obtained is called frequency polygon. It has more than four sides.

Polygon:

Marks	0-20	20-40	40-60	60-80	80-100	Total
No. of students(f)	7	10	20	8	5	50
Mid Values	10	30	50	70	90	----

Frequency Curve:

A frequency curve can be drawn through the various points of the polygon. The curve is drawn freehand in such a manner that the area included under the curve is

approximately the same as that of the polygon . The object is to eliminate as far as possible all accidental variations that might be present in the data.

Ogive or Cumulative Frequency Curve:

If the upper limits of the continuous class intervals are represented in X-axis and the cumulative frequencies are represented to the Y-axis, we get cumulative frequency polygon. A free hand curve to smooth a cumulative frequency polygon is called an Ogive. An ogive is always non-increasing but may be parallel to X-axis.

Marks	0-20	20-40	40-60	60-80	80-100	Total
No. of students(f)	7	10	20	8	5	50
Mid Values	10	30	50	70	90	----
Cumulative Frequency	7	17	37	45	50	----

Exercise 04:

Below is given the frequency distribution of daily wages of 100 workers in a factory:

Daily wages (in Tk.)	501-700	701-900	901-1100	1101-1300	1301-1500	1501-1700	1701-1900
No. of worker	5	15	21	25	18	10	6

Draw a frequency polygon, frequency curve, histogram and ogive.

Solution:

Daily wages (in Tk.)	501-700	701-900	901-1100	1101-1300	1301-1500	1501-1700	1701-1900
No. of worker	5	15	21	25	18	10	6
Real class interval	500.5-700.5	700.5-900.5	900.5-1100.5	1100.5-1300.5	1300.5-1500.5	1500.5-1700.5	1700.5-1900.5
Mid Values	600.5	800.5	1000.5	1200.5	1400.5	1600.5	1800.5

Rule for converting inclusive class interval to exclusive class interval: Difference of Upper limit and next class lower limit is 01. Difference will be divided by 2 and is found as 0.5. 0.5 will be deducted from lower limit and 0.5 will be added with upper limit.

Exercise 05:

The fuel consumption of 500 cars of a city are given below:

Fuel Consumption (in liter)	0-100	100-200	200-300	300-400	400-500
No. of cars	80	125	200	75	20

Draw a frequency polygon, frequency curve, histogram and ogive.

Diagram:

Difference Between Graphs and Diagrams

- All graphs are a diagram but not all diagrams are graph. This means that diagram is only a subset of graph.
- Graph is a representation of information using lines on two or three axes such as x, y, and z, whereas diagram is a simple pictorial representation of what a thing looks like or how it works.
- Graphs are representations to a scale whereas diagrams need not be to a scale
- Diagrams are more attractive to look at which is why they are used in publicity whereas graphs are for the use of statisticians and researchers.

Values of mean and median can be calculated through graphs which is not possible with diagrams

- Graphs are drawn on graph paper whereas diagrams do not need a graph paper
- For frequency distribution, only graphs are used and it cannot be represented through diagrams

Data can be mainly presented by diagram in two ways. Such as, a) Bar Diagram; b) Pie Diagram

Bar Diagram:

Bar diagram is drawn by erecting a series of blocks of equal width but with proportional to the values corresponding to different time periods or categories. The vertical blocks are alternatively known as bars and hence the name Bar Diagram.

Pie Diagram:

The pie diagram is intended to compare the distinct components, which together constitute a whole. The whole is represented by a circle of arbitrary radius and the segments of the circle represent the component parts.

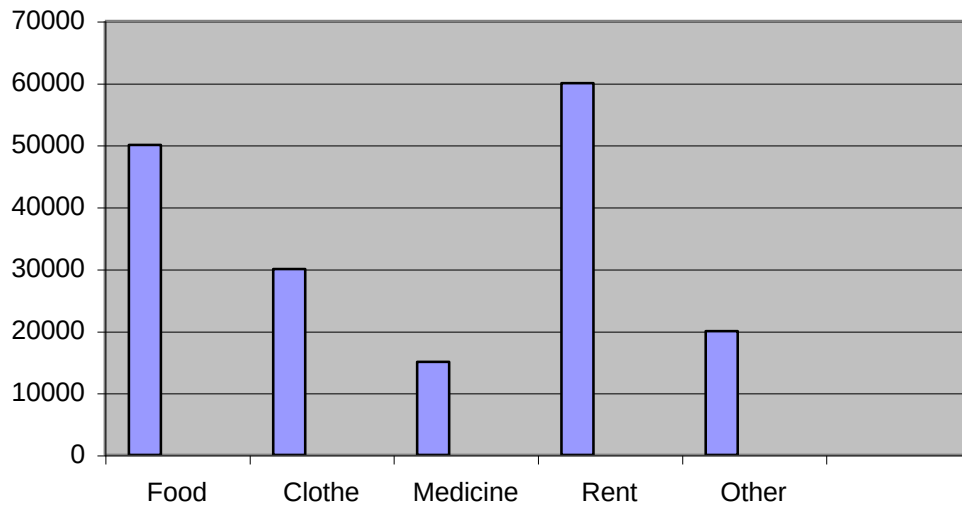
Example 03:

Expenditure amount (in Tk.) of different items of higher class family are given in the following table

Items	Food	Clothe	Medicine	Rent	Other
Amount (in Tk.)	50,000	30,000	15,000	60,000	20,000

Draw a bar diagram and pie diagram.

Solution:

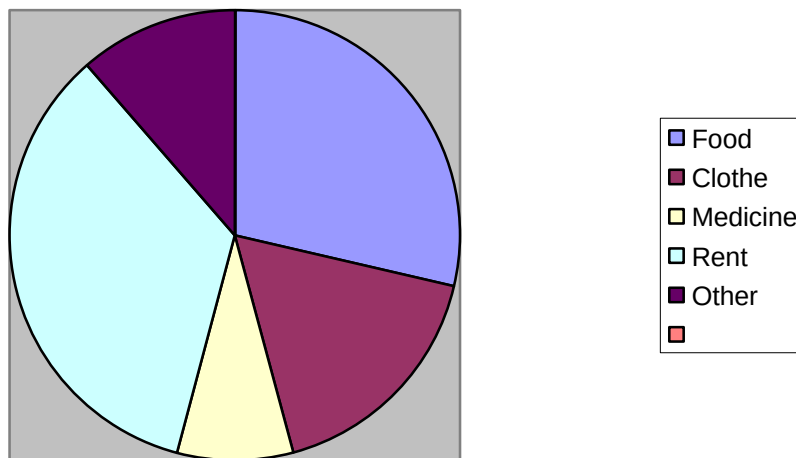


Bar Diagram

Table01: Cumulative amount of angle of different items

Items	Food	Clothe	Medicine	Rent	Other	Total
Amount (in Tk.) (f)	50,000	30,000	15,000	60,000	20,000	n=175,000
Amount of angle= $(f/n) \times 360$	103	62	31	123	41	360
Cumulative amount of angle	103	165	196	319	360	-----

(50000/175000)



Pie Diagram

Exercise 06:

Mr. Banna, cost analyst for a major firm, has compiled price data for four of the company's products. The figures are shown in the table:

Products	Dishwasher	Washing machine	Dryer	Refrigerators
Price (in Tk.)	2500	9000	2000	15500

Draw a bar diagram and Pie diagram.

Solution: